

Ravee Mishra

AI/ML Engineer | Computer Vision | Deep Learning | Generative AI
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AI/ML Engineer and M.Tech Computer Science student focused on computer vision, deep learning, generative AI, multimodal models, and robust intelligent systems. Experienced with PyTorch, transformer architectures, object detection, semantic segmentation, adversarial machine learning, and applied research workflows.

EDUCATION

M.Tech, Computer Science and Engineering — Indian Institute of Technology, Tirupati 2025 – Present | CGPA: 8.83

KIA Scholarship recipient for academic performance.

M.Sc. Mathematics — Indian Institute of Technology, Kanpur India

B.Sc. (Hons.) Mathematics and Statistics — University of Delhi India

EXPERIENCE

M.Tech Technical Project — Vision Tech Lab, IIT Tirupati 2026 – Present | IIT Tirupati

Working on multimodal models for applied computer vision and AI research.

Half-Time Teaching Assistant (HTTA) — IIT Tirupati 2025 – Present | IIT Tirupati

Supporting teaching and academic activities in the Computer Science and Engineering program.

SELECTED PROJECTS

Geospatial Semantic Segmentation 2026 | Research

Built a 4-channel Mask2Former pipeline for remote-sensing imagery that rasterizes annotations into semantic classes and exports georeferenced GeoTIFF predictions while preserving source alignment.

Technologies: PyTorch, Mask2Former, Transformers, rasterio, OpenCV, GeoTIFF

Channel Attention on YOLOv9 Research

Integrated CBAM-style channel attention into the YOLOv9 GELAN-C backbone and trained a Pascal VOC comparison pipeline with distributed training, confusion matrices, precision-recall curves, and timing analysis.

Technologies: PyTorch, YOLOv9, CBAM, Pascal VOC, torchrun

Jailbreaking Deep Models Academic project

Evaluated FGSM, multi-step PGD, and localized patch attacks on ImageNet classifiers, including transferability experiments from ResNet-34 to DenseNet-121. PGD reduced ResNet-34 Top-1 accuracy from 75.85% to 0.00%.

Technologies: PyTorch, torchvision, ResNet-34, DenseNet-121, FGSM, PGD

ADDITIONAL PROJECTS

GPT-2 (124M) in PyTorch

Academic project

Implemented the GPT-2 transformer architecture from scratch, loaded OpenAI's public checkpoint, and served the model through a Chainlit chat interface.

Technologies: PyTorch, Transformers, Chainlit

Few-Shot Writer Adaptation for HTR

2026 | Ongoing research

Re-implemented a multimodal in-context learning architecture for adapting handwriting recognition to a new writer using synthetic data and modest hardware.

Technologies: Python, PyTorch, multimodal learning

ACHIEVEMENTS

GATE (Data Science & Artificial Intelligence)

2025 — All India Rank 1023

GATE (Computer Science)

2025 — All India Rank 871 | Score: 731

CSIR NET (Mathematics)

2022 — All India Rank 111

IIT JAM (Mathematics)

2017 — All India Rank 85

IIT JAM (Statistics)

2017 — All India Rank 459

Innovation Recognition Award

*2026 — Ministry of Panchayati Raj
AI/ML Hackathon*

KIA Scholarship

2026 — Academic performance

TECHNICAL SKILLS

Programming & Frameworks: Python, C/C++, PyTorch

Deep Learning: CNN, ResNet50, VGG, channel attention, spatial attention

Computer Vision: Object detection, classification, segmentation, tracking, YOLOv9, Mask2Former

Transformers & LLMs: Self-attention, cross-attention, ViT, Swin Transformer, GPT-2

Generative AI: Autoencoders, VAE, GANs, diffusion models, Stable Diffusion